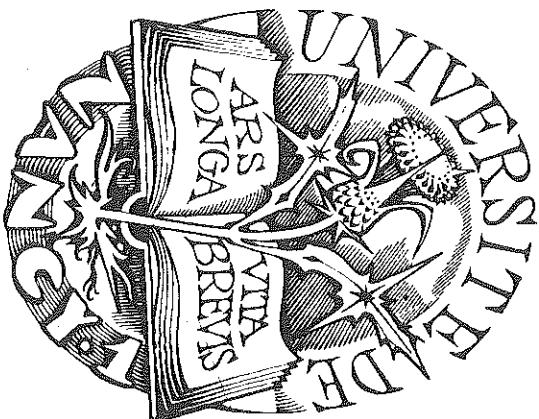




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Scalar curvature and isometry group

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SCALAR CURVATURE AND ISOMETRY GROUP

by

L. BERARD BERGERY

1. INTRODUCTION

Let M be a compact connected n -dimensional manifold ($n \geq 2$). For any Riemannian metric g on M , the scalar curvature u_g of g is a function on M (unless otherwise stated, manifolds, metrics, functions will be C^∞). A natural question is then, given the manifold M , which kind of functions we get as scalar curvatures? Much work has been done recently on this question, especially by J.L. Kazdan and F.W. Warner ([KW 1 to 4]), on the existence of positive scalar curvature by M. Gromov and H.B. Lawson ([GL 1 and 2]), R. Schoen and S.T. Yau ([SY 1 and 2]) and on the Yamabe conjecture by T. Aubin ([AU 1]). A recent review of all these results is [BB 1].

Here we are interested by the following generalization. We consider a compact Lie group G acting differentiably and effectively on M . For any G -invariant Riemannian metric g on M , the scalar curvature u_g is a G -invariant function on M . And the natural question is : which G -invariant functions we get as scalar curvatures of G -invariant Riemannian metrics?

Much of the preceding works have a straightforward generalization to the G -invariant case, but there are a few differences with the general case. This was already noticed by J.L. Kazdan and F.W. Warner ([KW 4]) for the simple example $M = S^2$, $G = S^1$, see §.2. In order to state the main results of the paper, we need the following

Definition :

Let G be a compact Lie group acting differentiably and effectively on the compact manifold M . Then the cohomogeneity of M under G , denoted $\text{coh}(M, G)$, is the codimension of the principal orbits of G in M .

We recall the following facts (see for example [BR] for precise definitions and statements). The principal orbits are in some sense the "greatest" ones ; the normal bundle to a principal orbit is trivial ; the union U of all the principal orbits is an open dense G -invariant submanifold of M ; the image U/G of U in the quotient space M/G is an open subset and a manifold of dimension precisely $\text{coh}(M, G)$.

Theorem A :

Suppose $n \geq 3$, $\text{coh}(M, G) \geq 2$ and G contains $SU(2)$ or $SO(3)$ as a subgroup. Then every G -invariant function on M is the scalar curvature of a G -invariant Riemannian metric.

We denote by G_0 the connected component of the identity in G . Remark that G does not contain $SU(2)$ or $SO(3)$ as a subgroup if and only if G_0 is abelian. In order to simplify the statement of the result in that case, we introduce some notations :

- $\mathcal{M}(M, G)$ is the set of all G -invariant Riemannian metrics on M ,
- $\mathcal{G}(M, G)$ is the set of all G -invariant functions on M .
- $\text{Scal}(M, G) \subset \mathcal{G}(M, G)$ is the set of the scalar curvatures of all G -invariant Riemannian metrics on M .]

Theorem B :

Suppose $n \geq 3$, $\text{coh}(M, G) \geq 2$, and G_0 abelian. Then we have one of the following three cases for (M, G) :

- a) there exists g in $\mathcal{M}(M, G)$ such that $u_g \geq 0$ and $u_g \neq 0$; then $\text{Scal}(M, G) = \mathcal{G}(M, G)$;
- b) there exists g in $\mathcal{M}(M, G)$ such that $u_g = 0$, but no g in $\mathcal{M}(M, G)$ has $u_g \geq 0$ and $u_g \neq 0$; then

$\text{Scal}(M, G) = \{ u \in \mathcal{G}(M, G) \mid u = 0 \text{ or } u < 0 \text{ somewhere on } M \}$; furthermore, if g in $\mathcal{G}(M, G)$ has $u_g = 0$, the Ricci curvature of g is identically zero.

- c) there is no g in $\mathcal{M}(M, G)$ with $u_g \geq 0$; then $\text{Scal}(M, G) = \{ u \in \mathcal{G}(M, G) \mid u < 0 \text{ somewhere on } M \}$.]

For the special case of dimension $n=2$, and $\text{coh}(M, G) = 2$ (that is, G finite) the result is very similar to the general case (see §.4).

The cases where $\text{coh}(M, G) = 0$ or 1 are considered in §.2 ; the results are quite different here. In §.3, we give some technical lemmas, adapted from the work of Kazdan and Warner. In §.5 we consider the generalization of Yamabe conjecture. In §.6, we prove the existence of invariant metrics with negative scalar curvature. Some more lemmas are considered in §.7 and the theorems A and B are proved in §.8.

Now, under the hypothesis of theorem B ($n \geq 3$, $\text{coh}(M, G) \geq 2$, G_0 abelian) a natural question is, given (M, G) to which class a) b) or c) it belongs ? Some examples show that the three classes are non void. The complete answer to this question is not known, even in the general case $G = \{1\}$. We have from one hand a construction of Riemannian metrics with positive scalar curvature by the techniques of surgery, and we indicate in §.11 the obvious generalization to the G -invariant case. On the other hand, there are obstructions to the existence of Riemannian metrics with positive scalar curvature.

Of course, if a manifold has no Riemannian metric with positive scalar curvature, it has no G -invariant Riemannian metric with positive scalar curvature, but the converse is not true, as we see on some examples in §.9, using the following

Theorem C :

Suppose that G_0 is abelian and all the orbits are principal. Then

M admits a G -invariant Riemannian metric with positive scalar curvature if and only if the quotient M/G (which is here a manifold) admits a Riemannian metric with positive scalar curvature. J

In §.10 we introduce a new obstruction for the existence of G -invariant Riemannian metric with positive scalar curvature when G_0 is abelian (and non zero).

Finally, in §.12 we show that these results give a complete answer to the question in the case $n=3$ and $G = S^1$.

2. LOW COHOMOGENEITY

Remark that $\text{coh}(M, G) = 0$ if and only if M is homogeneous under G . Then the scalar curvature of a G -invariant Riemannian metric on M is a constant, and the sign of this constant has been studied in [BB2] (in the general case of non necessarily compact homogeneous spaces). For comparison we recall here the results of [BB2] in the compact case.

Definition 2.1

Let M be a compact manifold, homogeneous under the compact Lie group G . Then M is normal if each G -invariant Riemannian metric on M is the quotient of a G -invariant Riemannian metric on G . (There are other characterizations of normal homogeneous spaces in [BB2]).

Theorem 2.2 (from [BB2])

Let M be a compact manifold, homogeneous under the compact Lie group G . Then :

- a) if M is normal and G_0 non abelian, $\text{Scal}(M, G)$ is the set of positive constant functions ;
- b) if G_0 is abelian, any G -invariant Riemannian metric on M is flat (and $\text{Scal}(M, G) = \{0\}$;
- c) in the other cases, $\text{Scal}(M, G) = \overline{\text{Scal}}(M, G)$ is the set of all constant functions. J

Remark 2.3

In [BB2], the group G is connected, but the proof is the same in the general case.

Now the case where $\text{coh}(M, G) = 1$ is completely different since there may be some "qualitative" obstruction. We recall here the following example, due to Kazdan and Warner [KW 4].

Let $M = S^2$ and $G = S^1$ acting by rotations around the N -S axis. Starting from N -pole, we may write any G -invariant Riemannian metric in polar coordinates :

$$g = dr^2 + f(r)^2 d\theta^2$$

where $2\pi f(r)$ is the length of the parallel at distance r from N -pole. If $2L$ is the length of a meridian, then

$$f(0) = f(L) = 0 \text{ and } f'(0) = 1 = -f'(L) .$$

Then the scalar curvature (which is twice the Gauss curvature) is a function of r only, given by :

$$u = -2 \frac{f''}{f'}$$

Proposition 2.4 (J.L. Kazdan and F.W. Warner [KW 4])

If u is monotone^{ic}, it is a positive constant. J

So there is a "qualitative" obstruction for a G -invariant function to be the scalar curvature of a G -invariant Riemannian metric. We will not study the case $\text{coh}(M, G) = 1$ here, because the general methods of the paper do not apply. We hope to come back to that case in a near future. In order to state the list of obstructions known (to the author) up to now, we need a

Definition 2.5 :

We will say that G acts irreducibly on M if the principal orbits admits only one G -invariant Riemannian metric, up to homothety.

Theorem 2.6 : Let $\text{coh}(M, G) = 1$.

- a) If $M = S^n$ and G acts irreducibly with principal orbit S^{n-1} , then if $u \in \text{Scal}(M, G)$, u is positive somewhere and if moreover u is monotonous, it is constant
- b) If $M \neq S^2$ or $\mathbb{R}P^2$ and G acts irreducibly with principal orbit a disjoint union of tori T^{n-1} , if $u \in \text{Scal}(M, G)$ then $u = 0$ or u is positive somewhere and negative somewhere on M (i.e. u takes effectively the two signs).
- c) In the other cases where G acts irreducibly, if $u \in \text{Scal}(M, G)$ then u is positive somewhere.
- d) If G does not act irreducibly and all orbits are disjoint union of tori T^{n-1} , if $u \in \text{Scal}(M, G)$, then $u = 0$ or u is negative somewhere. J

This theorem will be proved in a forthcoming paper on the curvature of Riemannian manifold of cohomogeneity one.

3. Some technical lemmas

In this paragraph, we generalize to the G -invariant case the main result of J.L. Kazdan and F.W. Warner ; the demonstrations are very closed to that of [KW 2 and 3], so they will be only sketched.

We denote $C^0(M, G)$ the set of continuous G -invariant functions on M , and $\|\cdot\|_p$ the L^p -norm on M associated with a Riemannian metric g .

Lemma 3.1 : (Equivalent approximation lemma)

Assume $\text{coh}(M, G) \geq 2$. Let $f, h \in C^0(M, G)$, then $\min f \leq h(x) \leq \max f$, $\forall x \in M$, if and only if, given any $\epsilon > 0$, there is a G -equivariant diffeomorphism α of M such that $\|f \circ \alpha - h\|_p < \epsilon$.

Proof :

The proof is the same than that of theorem 2.1 of [KW 2], but working on M/G instead of M . The only difficulty is that M/G is not necessarily a manifold. Notice that f and h induce continuous functions $\bar{f}$ and $\bar{h}$ on M/G such that $f = \bar{f} \circ \pi$ and $h = \bar{h} \circ \pi$.

With the notations of theorem 2.1 of [KW 2], we choose a triangulation Δ of M/G such that the image $M/G - U_\epsilon$ of non principal orbits be in the $(q-1)$ -skeleton of Δ , where $q = \dim M/G = \text{coh}(M, G) \geq 2$. Then we may do the construction of J.L. Kazdan and F.W. Warner in such a way that we avoid $M/G - U_\epsilon$. In particular we choose the diffeomorphism φ such that it is a diffeomorphism of U_ϵ which is the identity near the boundary of U_ϵ and is isotopic to the identity through such diffeomorphisms. Now there is a G -equivariant diffeomorphism α of M such that $\pi \circ \alpha = \varphi \circ \pi$, and the rest of the proof is trivial. J

Lemma 3.2 (Equivalent inverse function lemma)

Assume $\text{coh}(M, G) \geq 2$, $p > \dim M$ and $g \in \mathcal{M}(M, G)$. If u_g is not a nonnegative constant, there exists an $\eta > 0$ such that, if $f \in \mathcal{F}(M, G)$ and $\|f - u_g\|_p < \eta$, there exists $\tilde{g} \in \mathcal{M}(M, G)$ with $u_{\tilde{g}} = f$.

Proof :

With the same notations than in lemma 2 of [KW 3], let $U \subset H_{g,p}(M, \mathbb{R})$ be a G -invariant neighborhood of zero, so small that $Q : U \rightarrow Q(U) \subset L_p(M, \mathbb{R})$ be invertible. Choose η such that

$\|f - u\|_p < \eta$ implies $f \in Q(U)$. Since f, g and A^* are G -invariant,

if $u \in U$ is such that $Q(u) = f$ then $Q(u \cdot \alpha) = f$ for each α in G .

But $u \cdot \alpha$ is also in U where Q is invertible so $u = u \cdot \alpha$,

that is u is G -invariant and also $\tilde{g} = g + A^*u$. Finally $f = u_{\tilde{g}}$

and $\tilde{g}$ is smooth if f is smooth. $\square$

Proposition 3.3 :

Assume $\text{coh}(M, G) \geq 2$. Then

- a) if $\text{Scal}(M, g)$ contains the constant function 1, then it contains every f in $\mathcal{R}(M, G)$ which is positive somewhere ;
- b) if $\text{Scal}(M, g)$ contains the constant function 0, then it contains every f in $\mathcal{R}(M, G)$ which is positive somewhere and negative somewhere ;
- c) if $\text{Scal}(M, g)$ contains the constant function -1, then it contains every f in $\mathcal{R}(M, G)$ which is negative somewhere.

Proof :

We prove (a), the proofs for (b) and (c) are almost the same. Let g_1 in $\mathcal{M}(M, G)$ have scalar curvature $u_{g_1} = 1$. Of course, for every $c > 0$, $u_{cg_1} = \frac{1}{c}$. Suppose now f non constant. Since f

is positive somewhere, there exists $c > 0$ such that

$\min f < u_{cg_1} < \max f$. Since $\text{coh}(M, G) \geq 2$, there exists

$g_2 \in \mathcal{M}(M, G)$ near cg_1 such that $\min f < u_{g_2}(x) < \max f$.

$\forall x \in M$, and u_{g_2} is not a positive constant. Then by lemma 3.2,

there exists $\eta > 0$ such that $\|h - u_{g_2}\|_p < \eta$ for h in $\mathcal{R}(M, G)$

implies $h \in \text{Scal}(M, G)$. Now by lemma 3.1, there exists a G -equivariant diffeomorphism α of M such that $\|f \circ \alpha - u_{g_2}\|_p < \eta$, so $f \circ \alpha \in \text{Scal}(M, G)$, that is there exists g_3 in $\mathcal{M}(M, G)$ with $u_{g_3} = f \circ \alpha$. Then $f = u_g$ where $g = (\alpha^{-1})^* g_3$. $\square$

Remark :

It is possible to refine proposition 3.3 by fixing a globally conformal class of G -invariant metrics as in theorem 6.2 of [KW 2].

4. THE 2-DIMENSIONAL CASE (G finite)

In the particular case of dimension 2, we recall that Gauss-Bonnet formula

$$\int_M u_g dv_g = 4\pi \chi(M)$$

(where dv_g is the volume element associated to g and $\chi(M)$ the Euler characteristic of the surface M) gives a sign condition on u_g [KM 1].

Remark also that here $\text{coh}(M, G) = 2$ implies G finite. We obtain a result exactly similar to the case with trivial G .

Theorem 4.1:

Let M be a compact connected surface (i.e. $n=2$) and G a finite group acting effectively on M . Then

- a) if M is S^2 or $\mathbb{RP}^2$,
 $\text{Scal}(M, G) = \{u \in \mathcal{G}(M, G) \mid u > 0 \text{ somewhere on } M\}$
- b) if M is T^2 or K^2 (Klein bottle)
 $\text{Scal}(M, G) = \{u \in \mathcal{G}(M, G) \mid u \equiv 0 \text{ or } u > 0 \text{ somewhere and } u < 0 \text{ somewhere on } M\}$
- c) in the other cases (i.e. $\chi(M) < 0$)
 $\text{Scal}(M, G) = \{u \in (M, G) \mid u < 0 \text{ somewhere on } M\}$.

Proof:

The Gauss-Bonnet formula shows easily that the sign condition is necessary (remark that dv_g depends on g). In the other way, we know that M admits a G -invariant metric with constant scalar curvature

[> 0 in the case (a), 0 for (b) and < 0 for (c)]. Then the proposition 3.3 gives the sufficiency of the sign condition. $\square$

5. GENERALIZED YAMABE CONJECTURE

We suppose now $n \geq 3$. Then Yamabe conjecture states that for every Riemannian manifold (M, g) , there exists a Riemannian metric $\tilde{g}$ (pointwise) conformal to g with constant scalar curvature. In order to state more precisely the generalized conjecture in the G -invariant case, we introduce the following definition.

Let (M, g) be a n -dimensional compact connected Riemannian manifold and G a subgroup of the isometry group. We denote

$$\mathcal{G}^+(M, G) = \{f \in \mathcal{G}(M, G) \mid f > 0 \text{ everywhere on } M\}$$

For every f in $\mathcal{G}^+(M, G)$, we denote :

$$J_g(f) = \frac{4 \frac{n-1}{n-2} \int_M |df|^2 dv_g + \int_M u_g f^2 dv_g}{\left(\int_M f^{\frac{2n}{n-2}} dv_g \right)^{\frac{n-2}{n}}}$$

Definition 5.1

$$\mu_G(g) = \inf_{f \in \mathcal{G}^+(M, G)} J_g(f)$$

Remark that $J_g(f)$ is always bounded from below so that $\mu_G(g)$ is well defined for every G and g . Now we may state :

Generalized Yamabe conjecture 5.2

There exists a G -invariant Riemannian metric $\tilde{g}$ on M , (pointwise) conformal to g , such that $u_{\tilde{g}}$ is constant and equal to $\mu_G(g)$ and $\text{vol}(M, \tilde{g}) = 1$.

Proposition 5.3

The conjecture is equivalent to the existence of $f \in \mathcal{G}^+(M, G)$ such that $\mu_G(g) = J_g(f)$.

Proof:

The Euler formula for the functional J_g on $\mathcal{G}^+(M, G)$ gives if f realizes the infimum,

$$4 \frac{n-1}{n-2} \Delta_g f + u_g f = \lambda f^{\frac{n+2}{n-2}}$$

for some constant λ . We remark that $J_g(f)$ is invariant by homothety, so we may choose $\int_M f^{\frac{2n}{n-2}} dv_g = 1$.

Then, integrating the formula against f , we obtain $\lambda = J_g(f) = \mu_G(g)$.

Now the scalar curvature of $\tilde{g} = f^{\frac{4}{n-2}} g$ is given by

$$\mu_g = f^{\frac{n+2}{n-2}} (4 \frac{n-1}{n-2} \Delta_g f + u_g f) = \mu_{\tilde{g}}(g) \quad \text{and}$$

$$\text{Vol}(M, \tilde{g}) = \int_M f^{\frac{2n}{n-2}} dv_g = 1 \quad \square$$

When G is reduced to the identity, then $\mu_{\{1\}}(g)$ is usually denoted $\mu(g)$ and we have the usual Yamabe conjecture; it is almost completely solved in that case (H. Yamabe [YA], N.S. Trudinger [TR], T. Aubin [AU 1]) and moreover, if $\mu(g) \leq 0$ the solution is unique. We denote ω_n the volume of the canonical n -sphere.

Theorem 5.4

If $\mu_g(g) < n(n-1)\omega_n^{\frac{2}{n}}$, then generalized Yamabe conjecture is true for (M, g) and G .

Proof :

Suppose first $\mu_g(g) \leq 0$. Then obviously, if g is G -invariant, $J_g(f) = J_g(f \circ \varphi)$ for any φ in G . Since the solution f of Yamabe conjecture for $G = \{1\}$ is unique in that case, it is necessarily G -invariant and then $\mu_g(g) \leq J_g(f) = \mu(g)$. But obviously $\mu_g(g) \geq \mu(g)$ so $\mu_g(g) = J_g(f)$.

In the remaining case $\mu(g) > 0$, we have

$\mu(g) \leq \mu_g(g) < n(n-1)\omega_n^{\frac{2}{n}}$, so by Aubin there exists a solution, but it is not necessarily unique. Now the proof by Aubin may be translated to the G -invariant case by restricting every function space to the subspace of G -invariant functions. $\square$

6. Existence of metrics with negative scalar curvature :

We generalize to the G -invariant case a theorem of T. Aubin ([AU 2]).

Proposition 6.1 :

Let $n \geq 3$ and $\text{coh}(M, G) \geq 2$. Then for each $c > 0$ there exists on M a G -invariant Riemannian metric with constant negative scalar curvature $-c$ and volume 1.

Proof :

Due to the generalized Yamabe conjecture, it is sufficient to have a g in $\mathcal{M}(M, G)$ with $\mu_g(g) < 0$. Remark that

$$\mu_g(g) \leq \int_M u_g dv_g [\text{Vol}(M, G)]^{\frac{n}{n-2}}$$

so it is sufficient to have a g in $\mathcal{M}(M, G)$ with $\int_M u_g dv_g < \int_M \Delta g^{-1}$.

We suppose first that the dimension p of principal orbits is positive. Since the normal bundle to the principal orbit is trivial, there exists a neighborhood V of some principal orbit which is G -equivariantly diffeomorphic to $G/K \times D^q$ where G/K is isomorphic to a principal orbit, G acts on the left on G/K and D^q ($q = n-p$) is an open disk on which G acts trivially. We choose on $G/K \times D^q$ the Riemannian product metric $g_1 = g_2 + g_3$, where g_2 is a fixed G -invariant Riemannian metric on G/K and g_3 a fixed Riemannian metric on D^q . We pull back g_1 to V and there exists a G -invariant Riemannian metric $\bar{g}$ on M which coincides with g_1 on V .

We choose now a positive function f on D^q which is identically 1 near the boundary, and we let

$$g_4 = f^{\frac{2p}{n-1}} (f^2 g_2 + g_3).$$

Since f is constant near the boundary, the pull back of g_4 to V and $\bar{g}$ on $M-V$ agree to give a G -invariant Riemannian metric g on M .

$$\text{Now } \int_M u_g \, dv_g = \int_{M-Y} u_g \, dv_g + \int_{G/K \times D^q} u_{g_q} \, dv_{g_q} .$$

The first term does not depend on f . So it is sufficient to choose f such that the second term be negative with arbitrarily large absolute value.

An elementary calculation gives :

$$\int_{G/K \times K^q} u_{g_q} \, dv_{g_q} =$$

$$\text{Vol}(G/K, g_2) \int_{D^q} \frac{r^{n-1}}{r^{n-1}-2} \left[u_{g_2} + r^2 u_{g_3} - \frac{p(n-1-p)}{n-1} \|df\|^2 \right] dv_{g_3} .$$

Notice that u_{g_2} is a constant and that $\text{coh}(M, G) \geq 2$ implies $p \leq n-2$, so the coefficient of $\|df\|^2$ is negative.

Remark that

$$\text{Vol}(M, g) = \text{Vol}(M-Y, g) + \text{Vol}(G/K \times D^q, g_q) ,$$

and here also the first term does not depend on f , and

$$\text{Vol}(G/K \times D^q, g_q) = \text{Vol}(G/K, g_2) \cdot \int_{D^q} \frac{r^{n-1}}{r^{n-1}-2} dv_{g_3}$$

Now, if we choose g_3 such that u_{g_3} is a negative constant we may choose f as great as we want near the center of D and we get $\int_M u_g \, dv_g$ as small as we want, even keeping $\text{Vol}(M, g)$ bounded.

In the case where G is finite, we choose a disk D^n such that D^n is disjoint from its translates by G . Inside D^n we choose some "handle" $S^p \times D^q$ with $p+q = n$, $p \geq 1$ and $q \geq 2$ (recall that $n \geq 3$) ; and we perform the same construction as above in a G -invariant way on $S^p \times D^q$ and its translates by G . $\square$

Remark :

In the case where G is finite, we may also apply a G -invariant version of the proof given in [Au 2] for the case with $G = \{1\}$.

7. Some properties of μ_G

First, we recall that $\mu(g)$ is a conformal invariant since, if

$$\varphi > 0 \quad \int \frac{4}{\varphi^{n-2}} g \quad (f) = \int g(\varphi f) .$$

We deduce immediately :

Proposition 7.1 :

If $\varphi \in \mathcal{G}^+(M, G)$, then $\mu_G(\varphi g) = \mu_G(g)$. $\square$

A second property is

Proposition 7.2 :

The functional $\mu_G : \mathcal{M}(M, G) \rightarrow \mathbb{R}$ is continuous for the C^∞ -topology on $\mathcal{M}(M, G)$.

Proof :

Let g_t be a 1-parameter family of G -invariant Riemannian metrics on M , which is C^∞ in all variables. Then for each f in $\mathcal{G}^+(M, G)$, $\int g_t(f)$ is continuous in t and the infimum $\mu_G(g_t)$ is lower semi-continuous in t . In order to simplify the proof of upper semi-continuity, we prove first

Lemma 7.3 :

There exists a 1-parameter family $\bar{g}_t$ of G -invariant Riemannian metrics such that $\mu_G(\bar{g}_t) = \mu_G(g_t)$ and the volume element dv_{g_t} is equal to some fixed volume element dv for all t , with $\int_M dv = 1$.

Proof of the lemma : There exists a positive function $\varphi(t)$ such that $\varphi(t) g_t$ satisfies $\text{Vol}(M, \varphi(t) g_t) = 1$ for each t . Then, by the equivariant version of Moser's lemma, there exists a one-parameter family of G -equivariant diffeomorphisms α_t of M such that the volume element of

$\alpha_t^*(\varphi(t)g_t) = \bar{g}_t$ be a fixed volume element dv , which satisfies $\int_M dv = 1$.

Of course $\mu_G(\alpha_t^*(\varphi(t)g_t)) = \mu_G(\varphi(t)g_t) = \mu_G(g_t)$. $\square$

Proof of upper semi-continuity of μ_G .

From the lemma, we see that it is sufficient to prove upper semi-continuity in $t = 0$ for a family g_t such that $dv_{g_t} = dv$ for each t with

$$\int_M dv = 1. \text{ We consider now}$$

$$A = \{f \in C^+(M, G) \mid \int_M f^{\frac{2n}{n-2}} dv = 1\}$$

Then $\mu_G(g_t) = \inf_{f \in A} I_{g_t}(f)$

$$\text{where } I_{g_t}(f) = \frac{4}{n-2} \int_M \|df\|_{g_t}^2 dv + \int_M u_{g_t} f^2 dv.$$

For each $\varepsilon > 0$ and each t , there exists f_t in A such that

$$\mu_G(g_t) \leq I_{g_t}(f_t) \leq \mu_G(g_t) + \frac{\varepsilon}{3}.$$

We take t in some compact interval around 0, so $\|u_{g_t}\|$ is uniformly

bounded by some constant K . In particular

$$\mu_G(g_t) \leq I_{g_t}(1) = \int_M u_{g_t} dv \leq K,$$

and $I_{g_t}(f_t) \leq K + \frac{\varepsilon}{3}$

Now, there exists η such that $|t| < \eta$ implies $\|u_{g_t} - u_{g_0}\| \leq \frac{\varepsilon}{3}$ and for each 1-form ω on M ,

$$\left| \|\omega\|_{g_t}^2 - \|\omega\|_{g_0}^2 \right| \leq \|\omega\|_{g_0}^2 \frac{\varepsilon}{6K + \varepsilon}.$$

Then for such $|t| < \eta$,

$$\begin{aligned} |I_{g_t}(f_t) - I_{g_0}(f_t)| &\leq \frac{4}{n-2} \int_M \left| \|df_t\|_{g_t}^2 - \|df_t\|_{g_0}^2 \right| dv + \\ &\quad \int_M |u_{g_t} - u_{g_0}| f_t^2 dv \\ &\leq \frac{\varepsilon}{6K + \varepsilon} \frac{4}{n-2} \int_M \|df_t\|_{g_t}^2 dv + \frac{\varepsilon}{3} \int_M f_t^2 dv. \end{aligned}$$

Since $\frac{2}{n} + \frac{n-2}{n} = 1$, we have

$$\int_M f_t^2 dv \leq \left(\int_M dv \right)^{\frac{2}{n}} \left(\int_M f_t^{\frac{2n}{n-2}} dv \right)^{\frac{n-2}{n}} = 1.$$

Also $\frac{4}{n-2} \int_M \|df_t\|_{g_t}^2 dv = I_{g_t}(f_t) - \int_M u_{g_t} f_t^2 dv \leq K + \frac{\varepsilon}{3} + K$

Hence

$$|I_{g_t}(f_t) - I_{g_0}(f_t)| \leq \frac{2\varepsilon}{3}$$

Since $\mu_G(g_t) \geq I_{g_t}(f_t) - \frac{\varepsilon}{3}$ and $\mu_G(g_0) \leq I_{g_0}(f_t)$, we get finally:

$$\text{for each } \varepsilon > 0, \text{ there exists } \eta > 0 \text{ such that } |t| < \eta \text{ implies}$$

$$\mu_G(g_t) - \mu_G(g_0) \geq I_{g_t}(f_t) - \frac{\varepsilon}{3} - I_{g_0}(f_t) \geq -\varepsilon.$$

And this proves proposition 7.2. $\square$

I. Aubin has shown in [AU1] that $\mu(g)$ is always bounded by $n(n-1)\omega_n^{\frac{2}{n}}$. This is not true for $\mu_G(g)$ in full generality. For example we consider

$M = S^{n-1}$ with $G = Sp(n+1)$ acting transitively; then M admits a 1-parameter family g_t of G -invariant Riemannian metrics of volume 1.

There are homogeneous and it is known (see for example [BE]) that the scalar curvature u_{g_t} of that family is not bounded. Remark that here $\mu_G(g_t) = u_{g_t}$

since M is homogeneous and $\text{Vol}(M, g_t) = 1$. I suspect that $\mu_G(g)$ is bounded when $\text{coh}(M, G) \geq 2$, but we will not need it here.

8. Proof of theorems A and B

We prove first theorem B. Under the hypothesis $\text{coh}(M, G) \geq 2$ and $n \geq 3$, we have already proved in proposition 6.1 that there exists on M a G -invariant Riemannian metric g_1 with constant negative scalar curvature. Then proposition 3.3 c) gives theorem B c). In the same way the first part of theorem B b) follows from proposition 3.3 b) and c) and proposition 6.1. The second part of theorem B b) follows from the

Lemma 8.1 :

Assume $u_G(g) \leq 0$ for each g in $\mathcal{M}(M, G)$. Then, if g in $\mathcal{M}(M, G)$ satisfies $u_g = 0$, then its Ricci curvature satisfies $r_g = 0$.

Proof :

This is a generalization of a result of J.P. Bourguignon ([B0]), and the proof is the same ; if we consider the 1-parameter family $g_t = g + t r_g$, then for t sufficiently small $g_t \in \mathcal{M}(M, G)$ since r_g is G -invariant if g is G -invariant. And J.P. Bourguignon proved that for t sufficiently small, $u_{g_t} \geq 0$ and $\neq 0$. $\square$

Suppose now that there exists on M a G -invariant Riemannian metric g_2 with $u_{g_2} \geq 0$, $u_{g_2} \neq 0$; we see easily that $u_G(g_2)$ is positive.

We consider the 1-parameter family of G -invariant Riemannian metrics

$$g_t = (2-t)g_1 + (t-1)g_2, \quad t \in [1, 2]$$

with $u_G(g_1) < 0$. Since $u_G(g_t)$ is continuous in t , there exists a such that $u_G(g_a) = 0$ and by theorem 5.4 there exists $\tilde{g}_a$ such that $\tilde{u}_{\tilde{g}_a} = 0$. Also, there exists b such that $0 < u_G(g_b) < n(n-1) \omega_n^{\frac{2}{n}}$ and again by theorem 5.4, there exists $\tilde{g}_b$ such that $u_{\tilde{g}_b} = u_G(g_b) > 0$.

Now proposition 3.3 a) gives theorem B a). $\square$

Remark 8.2 :

The three classes in theorem B are non void in general. Theorem C gives many examples in the classes a) and c) and tori T^n are examples in class b). Notice that if (M, G) satisfies the hypothesis of theorem B b) and g has $u_g = 0$, then the Killing fields of the action of G are parallel, by the well known Bochner theorem, since $r_g = 0$. But M is not necessarily a product ; for example the Klein bottle has a flat Riemannian metric with an S^1 -action, but it is not a product. $\square$

Then we prove theorem A. Notice that in the above proof of theorem B, we did not use the hypothesis on G_0 so theorem A follows immediately from the preceding proof and from the

Theorem 8.3 : (H.B. Lawson and S.T. Yau [LY])

If a compact Lie group G acts differentiably and effectively on a compact manifold M , and G contains $SU(2)$ or $SO(3)$ as a subgroup, then there exists on M a G -invariant Riemannian metric with positive scalar curvature. $\square$

9. Proof of theorem C and examples :

Under the hypothesis of theorem C, the quotient space M/G is a manifold B and the projection $\pi : M \rightarrow B$ is a fiber bundle with structural group G and fibre a disjoint union of tori (since G_0 is abelian). Now, for any G -invariant Riemannian metric g on M , there exists one and only one Riemannian metric g^* on B such that π is a Riemannian submersion from (M, g) to (B, g^*) .

If G is finite, π is a covering and the theorem is trivial. We suppose now that $p = \dim G > 0$. Remark that p is also the dimension of the fibres.

Let A and T be 0'Neill tensors of that Riemannian submersion (see [0N]) for definitions and computations of curvatures), (X_i) and (U_α) local orthonormal basis of the horizontal and vertical distribution respectively. Then, if $N = \sum_\alpha T_{U_\alpha} U_\alpha$, we have the following relation between scalar curvatures

$$U_g = U_{g^*} \circ \pi - \sum_{i,j} \|A_{X_i} X_j\|^2 - \sum_{\alpha,\beta} \|T_{U_\alpha} U_\beta\|^2 - \|N\|^2 + 2 \sum_i \langle D_{X_i} N, X_i \rangle$$

Let f be the positive function on B defined by

$$f(x) = \text{Vol}(\pi^{-1}(x)), \quad g \upharpoonright \pi^{-1}(x)$$

With the G -invariance, a local computation expresses N in terms of f alone and we get

$$U_g = (U_{g^*} + 2 \frac{\Delta^* f}{f} + \frac{p-1}{p} \frac{\|df\|^2}{f^2}) \circ \pi - \sum_{i,j} \|A_{X_i} X_j\|^2 - \sum_{\alpha,\beta} \|T_{U_\alpha} U_\beta\|^2$$

Where f^0 is the "traceless" part of T , that is

$$T_{U_\alpha}^0 V = T_{U_\alpha} V - \frac{g(U_\alpha, V)}{p} N$$

and Δ^* is the Laplacian of g^* .

In particular, if we suppose $U_g > 0$, then

$$U_{g^*} + 2 \frac{\Delta^* f}{f} + \frac{p-1}{p} \frac{\|df\|^2}{f^2} > 0$$

If $\dim B = n-p = 1$ then $B = S^1$, $U_{g^*} = 0$.

We look at a maximum of f and we get $\Delta^* f \leq 0$, $df = 0$, so a contradiction: we cannot have $U_g > 0$ in that case.

If $\dim B = n-p \geq 2$, the scalar curvature of $\tilde{g} = f^{\frac{2}{n-p-1}} g^*$ is given by

$$U_{\tilde{g}} = f^{\frac{n-2}{n-p-1}} (U_{g^*} + 2 \frac{\Delta^* f}{f} + \frac{n-p}{n-p-1} \frac{\|df\|^2}{f^2})$$

so we have $U_{\tilde{g}} > 0$, since $\frac{n-p}{n-p-1} > \frac{p-1}{p}$.

Conversely, let g^* be a Riemannian metric on B with $U_{g^*} > 0$. Choose a principal connection on the principal G -bundle associated to π ; this gives on M a G -invariant horizontal distribution $\mathcal{H}$ which is complementary to the vertical distribution $\mathcal{V}$ tangent to the orbits of G .

If we choose now a G -invariant metric h on the typical fiber F (necessarily flat), there exists by a theorem of J. Vilms ([Vil]) for each $t > 0$ a unique metric g_t on M such that g_t is G -invariant, π is a Riemannian submersion from (M, g_t) to (B, g^*) , with totally geodesic fibre isometric to $(F, t h)$ and with $g_t(\mathcal{V}, \mathcal{H}) = 0$. 0'Neill formulas give here

$$U_{g_t} = U_{g^*} - \|A\|_{g_t}^2$$

Notice that $A : \mathcal{H} \times \mathcal{H} \rightarrow \mathcal{V}$ depends only of the distribution $\mathcal{H}$, so

$$\|A\|_{g_t}^2 = t \|A\|_{g_1}^2$$

In particular, if $U_{g^*} > 0$ and t sufficiently small we have $U_{g_t} > 0$ and the theorem is proved. $\square$

As a consequence of theorem C, we will give two examples of pairs (M, G) where M admits a Riemannian metric with positive scalar curvature, but no G -invariant Riemannian metric with positive scalar curvature.

Example 9.1
Let Σ^9 be the 9-dimensional exotic sphere such that $\Sigma^9 \# \Sigma^9 = S^9$ and $\alpha(\Sigma^9) \neq 0$ so that Σ^9 has no Riemannian metric with positive scalar curvature (see N. Hitchin [H1]).

Now $\mathbb{R}P^9$ is also spin and $\alpha(\mathbb{R}P^9) = 0$, so the connected sum $\mathbb{R}P^9 \# \Sigma^9$ is a spin manifold with $\alpha(\mathbb{R}P^9 \# \Sigma^9) \neq 0$, since α is additive, and $\mathbb{R}P^9 \# \Sigma^9$ has no Riemannian metric with positive scalar curvature. We consider the universal cover

$$M = \mathbb{R}P^9 \# \Sigma^9 = \Sigma^9 \# \Sigma^9 = S^9, \text{ which admits of course Riemannian metrics with positive scalar curvature. Let } G = \mathbb{Z}_2 \text{ acts on } S^9 \text{ as the}$$

deck transformation group of that covering. Then S^9 has no $\mathbb{Z}_2$ -invariant Riemannian metric with positive scalar curvature for that action.

Example 9.2 :

Let B be the "K3 surface" that is the unique 4-dimensional simply connected compact manifold admitting Kähler metrics with zero Ricci curvature for some complex structure. Since $\hat{A}(B) \neq 0$ and B is spin, one knows that B has no Riemannian metric with positive scalar curvature.

Let M be a S^1 -principal bundle over B whose characteristic class is non zero and primitive in $H^2(B, \mathbb{Z}) = \mathbb{Z}^{2,2}$. Then one sees easily that $\pi_1(M) = 0$. By a result of M. Gromov and H.B. Lawson ([GL 2]), since M is 5-dimensional, it admits a Riemannian metric with positive scalar curvature. But from theorem C we see that M has no S^1 -invariant Riemannian metric with positive scalar curvature for the principal S^1 -action with quotient B .

10. Obstructions to positive scalar curvature :

The examples in §.9 show that, if G_0 is abelian, there are some new obstructions to positive scalar curvature for G -invariant Riemannian metrics on M . Notice that M.F. Atiyah and F. Hirzebruch have shown in [AH] that if a spin manifold M has an effective spin S^1 -action, then $\hat{A}(M) = 0$, so the classical Lichnerowicz obstruction to positive scalar curvature vanishes in the case of S^1 -actions. Here we give a generalization of Lichnerowicz obstruction under some restrictions.

Hypothesis :

Let M be a compact spin manifold and G a compact Lie group acting differentiably and effectively on M , which respects the given spin structure on M . We suppose moreover that G_0 is abelian and that all the orbits have the same dimension $r (= \dim G_0)$, and that $\text{coh}(M, G) \geq 2$.

We consider the subbundle E of TM tangent to the orbits. Since they are union of tori, E is canonically trivialized once we have chosen a basis of the Lie algebra of G_0 .

Riemannian

Let g be a G -invariant metric on M and F the orthogonal subbundle to E in TM . Since M is spin and E trivial, F is spin. In particular there exists a bundle S of spinors on F ; that is, the bundle F is associated to a $SO(n-p)$ -principal bundle on M (with $n-p = \dim F$), which may be lifted to a $\text{Spin}(n-p)$ -bundle; then S is associated to that bundle for the spin representation. Now the group G lets E invariant so G acts (by bundle-isomorphisms) on F and also on S since G respects the spin structure. We consider the space $\mathcal{G}^G$ of G -invariant section of S , and

Spinors with quadratic map $M \rightarrow \mathbb{Q} = H/\mathbb{Q}$, $F = \mathbb{P}^1 \mathbb{Q}$ and $S = \mathbb{P}^1$ spin bundle of $\mathbb{Q}$;

we define the corresponding Dirac operator

$$P : \mathcal{G}^G \longrightarrow \mathcal{G}^G \quad \text{by} \\ P(s) = \sum_i X_i \cdot \nabla_{X_i} s$$

P is naturally identified $\mathcal{G} \rightarrow \mathcal{G}$ by the same formula, since the $\mathcal{G}$ is $\mathbb{R}$ -linearly isomorphic to $\mathcal{G}^G$ via the map $s \mapsto \hat{A}(s)$ in case of free action, that of trivial map, in $\hat{A}(\mathbb{Q})$.

where (X_i) is a local orthonormal basis for F , $\cdot$ is the Clifford multiplication and ∇ is the connection on S induced from the restriction of the Levi-Civita connection to F .

Now let (u_α) be a basis of the Lie algebra of G , viewed as a subalgebra of vector fields on M . We consider the G -invariant function on M

$$f = \sqrt{\det(g(u_\alpha, u_\beta))}.$$

Then we have

Theorem 10.1 :

If the scalar curvature of g_0 is non negative and non identically zero, then the Dirac operator P on $\mathcal{G}^G$ associated to

$$g = f^{\frac{1}{2}} g_0 \quad \text{admits no nonzero solution of } Ps = 0.$$

Proof :

A straight-forward computation gives a formula (similar to Lichnerowicz's formula for the case without G) for the square of P associated to g

$$P^2 = C + \frac{k}{4}$$

$$\text{where } C = - \sum_i (\nabla_{X_i} \nabla_{X_i} - \nabla_{\nabla_{X_i} X_i})$$

(and ∇ is the connection induced from the Levi-Civita connection D of g), and k is the multiplication by the function on M

$$k = g + \sum_{i,j} \|A_{X_i} X_j\|^2 + \sum_{\alpha,\beta} \|V_\alpha V_\beta\|^2 - 2 \frac{\Delta f}{f} - \frac{3 \|\text{d}f\|^2}{f^2}$$

where (V_α) is a local orthonormal basis of E .

Here $A : F \times F \rightarrow E$ and $T : E \times E \rightarrow F$ are tensors defined in a way similar to O'Neill tensors of a submersion by

$$\langle A_X Y, U \rangle = \langle D_X Y, U \rangle$$

$$\text{and } \langle T_U V, X \rangle = \langle D_U V, X \rangle$$

where $X, Y \in F$, $U, V \in E$.

Notice that norms $\| \cdot \|$ are taken with respect to g and that Δ is the Laplacian of g .

$$u_g = f^{-\frac{n-2}{n-1}} u + 2 \frac{\Delta f}{f} + (3 - \frac{1}{n-1}) \frac{\|df\|^2}{f^2}$$

We denote T^0 the traceless part of T . We have

$$T_U^0 V = T_U V - \frac{\langle U, V \rangle}{p} Df,$$

So we get

$$k = f^{-\frac{n-2}{n-1}} u_g + \sum_{ij} \|A_{ij} X_j\|^2 + \sum_{\alpha\beta} \|T_{\alpha}^0 V_{\beta}\|^2 + (\frac{1}{p} - \frac{1}{n-1}) \frac{\|df\|^2}{f^2}$$

Since $p \leq n-2$, we have $k \geq 0$ and $k \neq 0$. Now P is selfadjoint for the scalar product

$$(s, s') = \int_M g(s, s') f^{-1} dv_g$$

and $C = \nabla^* \nabla$ where ∇^* is the adjoint of ∇ ($g_g \rightarrow g_g \otimes g^*$) for that scalar product.

Finally

$$(P_s, P_s) = (\nabla_s, \nabla_s) + \int_M k \|s\|^2 f^{-1} dv_g$$

and this second term is positive if $s \neq 0$, so $P_s \neq 0$.

Remark that P is an elliptic operator. For p even, s splits canonically in $S^+ \oplus S^-$ and correspondingly $g_g = g_g^+ \oplus g_g^-$, with

$$P : g_g^+ \rightarrow g_g^+ \cdot$$

This P has an index

$$\text{ind}(P) = \text{Ker}(P) \uparrow g_g^+ - \text{Ker}(P) \uparrow g_g^-$$

in $KG(\text{point}) = R(G)$, and $\text{ind}(P)$ depends only of the spin structure and the G -action.

Corollary 10.2 :

If M admits a G -invariant Riemannian metric with nonnegative nonzero scalar curvature, then $\text{ind}(P) = 0$ in $R(G)$.

Remark 10.3 : It is possible to generalize this invariant is the same way $\hat{A}(M)$ has been generalized in the case without G . We hope to come back to that in a near future.

11. A construction of metrics with positive scalar curvature

Let M_1 and M_2 be n -dimensional compact manifolds and suppose that the compact Lie group G acts differentiably and effectively on M_1 and M_2 . Let $N_1 \subset M_1$ and $N_2 \subset M_2$ be compact G -invariant submanifolds with normal bundle ν_1 and ν_2 respectively, and suppose that there exists a G equivariant bundle isomorphism φ from ν_1 to ν_2 . Let $D(\nu_1)$ and $D(\nu_2)$ be tubular neighbourhood of N_1 and N_2 such that φ induces a diffeomorphism from the boundary $S(\nu_1)$ of $D(\nu_1)$ to the boundary $S(\nu_2)$ of $D(\nu_2)$.

Let $M = (M_1 - D(\nu_1)) \cup_{\varphi} (M_2 - D(\nu_2))$ be the manifold obtained by gluing $M_1 - D(\nu_1)$ and $M_2 - D(\nu_2)$ by the diffeomorphism φ between their boundaries $S(\nu_1)$ and $S(\nu_2)$. Obviously, there is an action of G on M whose restrictions to $M_1 - D(\nu_1)$ and $M_2 - D(\nu_2)$ are the given actions.

Theorem 11.1 :

If M_1 and M_2 admit G -invariant Riemannian metrics with positive scalar curvatures and if the codimension of N_1 in M_1 (and N_2 in M_2) is ≥ 3 , then M admits a G -invariant Riemannian metric with positive scalar curvature.

Proof :

R. Schoen and S.T. Yau proved this theorem in the case where $G = \{1\}$ in [SY 2]. One sees easily that their proof works as well with a group G acting, since they used mainly the distance to N_1 in M_1 (and to N_2 in M_2) which is clearly G -invariant. Remark that the different proof given by M. Gromov and H.B. Lawson in the case of surgeries ([GL 2]) may also be adapted to obtain theorem 11.1.

Remark :

Gromov and Lawson have many applications of this result in the case $G = \{1\}$, using surgery and Smale's theory. But there is not such powerful results in the general case of a group action.

12. The 3-dimensional case with $G = S^1$

As an example, we consider the case where M is three-dimensional and $G = S^1$.

Theorem 12.1 :

Let $n = 3$ and $G = S^1$. Then

- a) if the action has fixed points, M admits S^1 -invariant Riemannian metrics with positive scalar curvature.
- b) if the action has no fixed point we have three cases :
 - 1) M is a classical finite quotient of S^3 or $S^2 \times S^1$ and admits S^1 -invariant Riemannian metrics with positive scalar curvature ;
 - 2) M is a finite quotient of T^3 and admits S^1 -invariant flat Riemannian metrics, but no Riemannian metric with positive scalar curvature ;
 - 3) in the other cases, M is a $K(\pi, 1)$ with $Z(\pi) = \mathbb{Z}$ and it admits no Riemannian metric with nonnegative scalar curvature.

Proof :

The S^1 action on 3-manifolds were completely classified by P. Orlik and F. Raymond (see [OR]). In the case b) the classification is a part of the classification of Seifert manifolds. Notice that R. Schoen and S.T. Yau have shown in [SY1] that a 3-dimensional $K(\pi, 1)$ admits no Riemannian metric with positive scalar curvature. Furthermore, a 3-dimensional manifold with zero Ricci curvature is flat, and, if compact, a finite quotient of T^3 .

In the case a) the classification by P. Orlik and F. Raymond shows essentially that M may be obtained by S^1 -equivariant connected sum along fixed points (of codimension 3 of course) of the manifolds $S^2 \times S^1$, $\mathbb{R}P^2 \times S^1$ and lens spaces $S^3/\mathbb{Z}_n$ with classical S^1 -action with fixed points, together with a manifold (noted $M_{0,0,0}$ in their classification) which may be obtained from $S^2 \times S^1$ with S^1 acting in the first factor by a S^1 -equivariant surgery along two fixed points, one in each of the two connected components of fixed points. A repeated application of theorem 11.1 gives theorem 12.1 a). $\square$

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